Corrective Actions Taken to Address the concerns in last visit

Criteria	Observation	Action Taken
1.PEOs	The process for assessing the achievement of Program Educational Objectives (PEOs) is clear. Key Performance Indicators (KPIs) are well-defined, and assessment tools are in use. However, there is a noticeable lack of correlation in the survey forms used to assess certain sub-KPIs related to PEO-2 and PEO-3. The cumulative KPI attainment process is well-documented in the approved policy.	The observation regarding the lack of correlation between certain survey items and the sub-KPIs of PEO-2 and PEO-3 is acknowledged. On review of the existing alumni and employer survey forms, the Department confirmed this concern and further noted that some questions were vaguely framed and that the PEOs were assessed through an uneven number of questions. Accordingly, the Department undertook a review activity and comprehensively revised both the alumni and employer survey forms. In the revised questionnaires, each question is directly aligned to a specific PEO, ambiguity has been removed, and every PEO is now measured through an equal number of questions to ensure balanced and valid attainment. The revised survey forms with clear PEO-to-question mapping are placed in the supporting record for verification.
	The Continuous Quality Improvement (CQI) process at the Program Educational Objectives (PEO) level has been initiated. Some assessment data has been collected using existing survey forms, and analysis and evaluation have been conducted. Stakeholder feedback is also gathered for the revision of survey forms. Corrective actions have been identified, but there are issues	The observation regarding the implementation of corrective actions at the Program Educational Objectives level is acknowledged. The Department has established and implemented a structured, evidence-based Continuous Quality Improvement process in which Program Educational Objectives are periodically evaluated using multiple stakeholder feedback mechanisms, and the findings are translated into documented corrective actions. Evidence of implementation includes the revision and formal approval of the Program Educational Objective evaluation system with improved assessment instruments and attainment methodology, the implementation of multiple assessment cycles, and the systematic review of results. In addition, recommendations arising from Industrial Advisory Board meetings have resulted in documented improvements, including the introduction of new elective courses. These actions demonstrate that the Department not only identifies

	with the implementation of these actions.	areas for improvement but also implements, monitors, and documents corrective actions as part of its Continuous Quality Improvement process.
2. PLOs	The department has updated the Program Educational Outcomes (PEOs), but it appears that some essential elements of PEO-2 are not addressed in the Program Learning Outcomes (PLOs).	The department appreciates the evaluators' observation regarding the alignment of PEO-2 with the Program Learning Outcomes. Upon review, the department identified that while PLO-8, PLO-9, and PLO-10 adequately address the leadership, teamwork, communication, and project management aspects of PEO-2, the element of developing innovative solutions was not explicitly reflected in the mapping. To address this gap, PLO-3 (Design/Development of Solutions) has been added to the mapping for PEO-2. PLO-3 emphasizes graduates' ability to design solutions for complex engineering problems and develop systems, components, or processes that satisfy specified requirements while considering societal, environmental, and safety constraints. This outcome directly supports graduates' capability to develop innovative software solutions. Accordingly, PEO-2 is now supported by PLO-3, PLO-8, PLO-9, and PLO-10, ensuring that all aspects of leadership, collaboration, effective communication, project management, and innovative solution development are comprehensively aligned.
	The course-to-PLO mapping demonstrates satisfactory alignment with all PLOs. Nevertheless, there are notable weaknesses in the mapping, particularly concerning PLO-3 and PLO-1, in terms of the learning domain and taxonomy level. However, the	The department acknowledges the evaluators' observations regarding the course-to-PLO mapping, particularly the alignment of PLO-1 and PLO-3 with the appropriate learning domains and Bloom's Taxonomy levels. As part of the department's established Continuous Quality Improvement (CQI) process, a comprehensive review of all course specifications is conducted after each semester and prior

	department has provided data regarding the ongoing mapping revision for upcoming semesters, which is based on Continuous Quality Improvement (CQI) and utilizes the course review performa.	to the commencement of the subsequent semester. Based on feedback from Course Review Reports, faculty discussions, and DBS recommendations, the Course Learning Outcomes (CLOs) of several courses have been revised to improve their wording, ensure measurable learning outcomes, and align the assigned cognitive domains and Bloom's Taxonomy levels with the intended Program Learning Outcomes. These revisions have resulted in an improved course-to-PLO mapping, particularly for PLO-1 (Engineering Knowledge) and PLO-3 (Design/Development of Solutions), ensuring that the CLOs reflect the appropriate cognitive complexity and learning domains required for the attainment of these outcomes. The revised CLOs, updated Bloom's Taxonomy levels, learning domains, and course-to-PLO mappings have been implemented for the current curriculum and are presented in Criterion 2 of the Self-Assessment Report as evidence of the department's Continuous Quality Improvement process.
	Assessment methodologies, including both direct and indirect methods, are established. However, the assessment methods for achieving PLOs through indirect means need improvement, especially in the case of the poorly designed internship form and the graduate exit survey forms. Furthermore, although direct assessment methods are in use, they are not consistently applied in certain	The department has comprehensively revised both the Internship Employer Feedback Form and the Graduate Exit Survey Form to improve their structure, clarity, and alignment with the Program Learning Outcomes. The revised instruments incorporate clearly defined assessment indicators that enable more reliable and meaningful indirect assessment of graduate attainment. To further strengthen the assessment process, the collection of employer feedback has been institutionalized through a dedicated Online Internship Feedback Module integrated with the department's existing Thesis Management System (TMS). This system facilitates systematic collection, storage, and analysis of employer evaluations for every internship. It also supports

	instances for a couple of PLOs, specifically PLO-8 and PLO-10.	the generation of consolidated, batch-wise reports with complete employer responses, thereby enhancing data reliability, traceability, and evidence-based Continuous Quality Improvement (CQI). Furthermore, the department has reviewed and updated the direct assessment methodology to ensure that all Program Learning Outcomes, including PLO-8 (Individual and Collaborative Teamwork) and PLO-10 (Project Management and Finance), are consistently and explicitly assessed through appropriate Course Learning Outcomes (CLOs) and assessment instruments. The revised assessment mappings have been incorporated into the relevant course specifications and also available in criteria 2.
	The Key Performance Indicators (KPIs) are clearly defined. However, there is an issue with the assessment process in direct assessment, as it is not consistently conducted at the appropriate taxonomy level. Additionally, the quality of the assessment process in the case of indirect methods appears to be subpar, especially when using the internship feedback form.	In response to the concern related to direct assessment, the department has undertaken a comprehensive review of all course specifications as part of its Continuous Quality Improvement (CQI) process. The Course Learning Outcomes (CLOs), their associated Bloom's Taxonomy (BT) levels, and the corresponding assessment instruments have been revised to ensure that student assessments are conducted at the intended cognitive level and are appropriately aligned with the relevant Program Learning Outcomes (PLOs). These revisions have been incorporated into the updated course files and are consistently implemented in subsequent course offerings. To address the concerns regarding indirect assessment, particularly the quality of the internship feedback process, the department has comprehensively redesigned the Internship Employer Feedback Form to improve its structure, assessment indicators, and alignment with the Program Learning Outcomes. Furthermore, the internship feedback process has been institutionalized through a dedicated Online Internship Feedback Module integrated with the department's existing Thesis Management System (TMS). The system enables systematic collection of employer evaluations, maintains complete assessment records, and

		generates consolidated batch-wise reports, thereby improving the reliability, consistency, and traceability of indirect assessment data.
3.Curriculum & Learning Process	The department has supplied a list of courses that cover CEP attributes. The visitation team has been presented with data related to CEPs and their assessments. However, the visitation team has observed a lack of understanding in the assessment of CEPs, along with identified weaknesses concerning the clarity and complexity level of CEP attributes in statements provided, particularly in a few courses.	The department acknowledges the observation made by the visitation team. We appreciate the valuable feedback regarding the understanding and assessment of CEP attributes, as well as the concerns about the clarity and complexity level of CEP statements in a few courses. To address this observation, the department has organized dedicated training sessions for all faculty members to improve their understanding of CEP formulation, mapping, and assessment in accordance with PEC guidelines. In addition, a standard CEP template, along with illustrative examples, has been prepared and shared with all course instructors.
	The use of rubrics-based assessment mechanisms in the lab is evident, as observed from the lab folders and during the visit to the labs. However, it is noted that the assessment mechanisms, especially in the Psychomotor and Affective domains, are not appropriate in certain lab courses, such as operating systems and deep learning. The visitation team also observed inconsistencies in the experiments included in the lab folders of a few courses, like Computer Architecture and Logic Design, where the	The department acknowledges the observation regarding the assessment mechanisms and laboratory course content. In response, the department has reviewed the assessment strategies employed in laboratory courses, with particular emphasis on the evaluation of the Psychomotor and Affective domains, to ensure their appropriateness and alignment with the intended learning outcomes. Furthermore, the department has streamlined the laboratory content in accordance with the approved course outlines and syllabi, ensuring that all experiments are relevant, consistent, and contribute effectively to the achievement of the defined learning outcomes. Assessment of the Affective and Psychomotor domains is conducted through rubric-based evaluations. These rubrics have been reviewed and approved by the KGH, ensuring that the laboratory activities are appropriately aligned with the prescribed outcomes and that student performance is evaluated consistently in accordance with the relevant BT levels.

	experiments did not align with the course contents as per the syllabus.	
	Course Learning Outcomes (CLOs) are defined for all courses, but the visitation team has identified issues with the taxonomy levels and their mapping to relevant Program Learning Outcomes (PLOs) in many course outlines. This discrepancy is notably observed in courses like Design and Analysis of Algorithms and Operating Systems etc.	The department acknowledges the evaluators' observations regarding the definition of CLOs, particularly the taxonomy levels and their alignment with the relevant PLOs in certain course outlines, including Design and Analysis of Algorithms and Operating Systems. As part of the department's established CQI process, a comprehensive review of all course specifications is conducted after each semester and prior to the commencement of the subsequent semester. Based on findings from Course Review Reports, faculty deliberations, and DBS recommendations, the CLOs of several courses have been revised to improve their clarity, ensure that they are measurable, and align the assigned learning domains and Bloom's Taxonomy levels with the intended Program Learning Outcomes. Particular attention has been given to ensuring that the CLOs are mapped to the appropriate cognitive levels and learning domains, resulting in improved course-to-PLO alignment, especially for PLO-1 (Engineering Knowledge) and PLO-3 (Design/Development of Solutions). These revisions ensure that the CLOs accurately reflect the expected cognitive complexity and support the effective attainment of the corresponding PLOs. The revised CLOs, updated Bloom's Taxonomy levels, learning domains, and course-to-PLO mappings have been implemented in the current curriculum. The updated course specifications and mapping matrices are presented in Criterion 2 of the Self-Assessment Report as evidence of the department's commitment to Continuous Quality Improvement and ongoing enhancement of curriculum quality.

	Assessment in the knowledge domain is generally appropriate, but at times it lacks rigor, as evident from the course folders, particularly in courses like Operating System, Organizational Behavior, and Computer Architecture and Logic Design. The assessments in the Psychomotor and Affective domains also require significant improvement for the evaluation of Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs). Additionally, it was noted that rubrics are not consistently followed for assessment in the Affective domain in some courses, specifically those mapped to PLO-10.	The department acknowledges the evaluators' observations regarding the level of rigor in the assessment of the knowledge domain in certain courses. The department maintains that its assessment system is designed to comprehensively evaluate student learning through a balanced combination of quizzes, assignments, laboratory work, projects, mid-semester examinations, and final examinations. For all courses, assessment instruments are developed in accordance with the defined CLOs and the assigned BT levels. The assessment components collectively cover a range of cognitive levels, ensuring that students are evaluated not only on knowledge recall and understanding but also on higher-order cognitive skills such as application, analysis, evaluation, and, where applicable, design and creation. The assessment distribution is reviewed during course planning and moderation to ensure adequate coverage of the intended learning outcomes and Bloom's Taxonomy domains. The department considers the assessment instruments used in courses such as Operating Systems, Organizational Behavior, and Computer Architecture and Logic Design to be appropriately aligned with the expected cognitive levels and sufficiently rigorous to assess the intended knowledge domain outcomes. Nevertheless, as part of the department's CQI process, assessment instruments are regularly reviewed through Course Review Reports, faculty discussions, and DBS meetings to identify opportunities for further enhancement. Feedback from internal and external stakeholders, including accreditation evaluators, is also incorporated to continuously strengthen the quality, consistency, and rigor of assessment practices across all courses.

	Satisfactory attainment is observed in all three domains. However, the visitation team noted that in several courses (e.g., Design and Analysis of Algorithms, Software Design and Architecture, Formal Methods, Internet of Things, etc.), assessments for Course Learning Outcomes (CLOs) are conducted at a lower taxonomy level. This should be addressed through Continuous Quality Improvement (CQI). It is also worth mentioning that despite the lower taxonomy levels, the assessment appears to be at a suitable level of complexity, indicating a potential lack of faculty understanding about taxonomy levels. This is an area that requires improvement.	The department acknowledges the evaluators' observations regarding the taxonomy levels assigned to the CLOs in courses such as Design and Analysis of Algorithms, Software Design and Architecture, Formal Methods, and Internet of Things. The department would like to clarify that these courses have been reviewed as part of the established CQI process, which is conducted after the completion of each semester. Based on the findings of the Course Review Reports, faculty deliberations, and DBS recommendations, the CLOs, their associated Bloom's Taxonomy levels, and the course-to-PLO mappings have been revised where necessary. As a result, the CLOs for these courses are now mapped to the appropriate higher cognitive levels, accurately reflecting the complexity of the intended learning outcomes and the assessment tasks. The department maintains that the assessment instruments including quizzes, assignments, laboratory work, projects, mid-semester examinations, and final examinations have consistently assessed students at the appropriate level of cognitive complexity. The revisions undertaken through the CQI process ensure that the documented Bloom's Taxonomy levels are now fully aligned with the rigor of the assessments and the intended learning outcomes. In addition, the department has reinforced faculty awareness and understanding of Bloom's Taxonomy through curriculum review meetings and internal discussions to promote consistent application of taxonomy levels during course design and assessment planning. The revised CLOs, updated taxonomy levels, and aligned assessment plans have been incorporated into the current curriculum and are presented in the SAR as evidence of the department's commitment to CQI and the ongoing enhancement of teaching, learning, and assessment practices

Faculty & Support Staff	There is a formal training mechanism in place for faculty development; however, it has been noted that this mechanism does not comprehensively cover all areas, particularly those related to learning domains and taxonomy levels.	The concern is acknowledged. The Department has a structured and comprehensive faculty development mechanism that explicitly includes training on Outcome-Based Education (OBE), covering the formulation of Course Learning Outcomes (CLOs), selection of appropriate learning domains, application of Bloom's Taxonomy levels, and mapping of CLOs to Program Learning Outcomes (PLOs). All newly inducted faculty undergo a dedicated one-week OBE training at the NUST Professional Development Centre (PDC), while regular refresher sessions and workshops are conducted at the department and SEECS levels to reinforce these concepts. During these training sessions, faculty are guided in developing measurable CLOs using the cognitive, psychomotor, and affective learning domains, selecting appropriate taxonomy levels, designing aligned assessment instruments, and ensuring constructive alignment between teaching, learning, and assessment. These practices are consistently implemented during course design, review, and OBE assessment activities. In addition to OBE-specific training, faculty regularly participate in national and international professional development programs to enhance their pedagogical and technical competencies. Therefore, the existing faculty development mechanism comprehensively addresses learning domains, taxonomy levels, and other OBE requirements through dedicated training, mentoring, and continuous professional development activities.
	Most of the teaching staff in the department are in the early stages of their academic careers, with a significant number being junior faculty members.	The faculty profile reflects a balanced mix of experienced senior academics and energetic early-career faculty. The department includes highly experienced faculty members with 19–20 years of experience along with extensive industry exposure who provide academic leadership, mentoring, curriculum development, research supervision, and quality assurance. To ensure effective academic delivery, all newly inducted faculty undergo structured induction and mentoring programs, including Outcome-Based Education (OBE) training, supervised teaching, and continuous professional development through the Department,

		SEECS, and the NUST Professional Development Centre (PDC). Junior faculty members are paired with experienced mentors during their initial semesters to ensure adherence to established teaching, assessment, and quality assurance practices. Choon choon na karain Therefore, although the Department has strengthened its faculty through the induction of promising early-career academics, they are supported by experienced faculty members and a robust mentoring framework. This combination provides continuity, innovation, and sustained academic quality in teaching, research, and student supervision.
Facilities and Infrastructure	The library has successfully acquired a substantial collection of books; however, it is currently facing budget-related constraints that need to be addressed in order to accommodate future growth. In addition to traditional resources, digital library resources are also available. It's worth noting that the library software is accessible, offering the potential for efficient resource management and organization once access issues are resolved. Efforts to expand physical space, address system-related challenges, and optimize the library software will greatly enhance the overall library experience for students and faculty, enabling seamless access to a wide range of educational resources.	The NUST Library System has undertaken several measures to strengthen library facilities, digital services, and access to educational resources. The Central Library currently provides seating for over 570 users, with additional seating arrangements during examination periods, four group discussion rooms, a seminar hall, and extended operating hours throughout the week. To improve service reliability, continuous coordination is maintained with the ICT Directorate for network enhancement, while regular maintenance and upgrades of the IT infrastructure ensure uninterrupted access to electronic resources. The library management systems, KOHA (Integrated Library Management System) and DSpace (Institutional Repository), are periodically upgraded and customized to enhance functionality, usability, security, and resource discovery. Students and faculty are provided access to a comprehensive collection of print and electronic resources, including books, journals, e-books, theses, and the HEC Digital Library, with remote access to major scholarly databases such as IEEE, ScienceDirect, SpringerLink, Wiley, Taylor & Francis, JSTOR, Emerald, ACS, AIP, and Scopus. In addition, the Institutional Repository is continuously updated with research publications, conference papers, and postgraduate theses. To further enhance library services, future initiatives include expansion of seating capacity, improved study and collaborative learning spaces, dedicated research commons and carrels, establishment of prayer areas and a

		coffee kiosk, and development of an AI-powered library chatbot to improve resource discovery and user support.
8. CQI	The CQI (Continuous Quality Improvement) process is in place and being practiced at all levels. However, the actual implementation of the CQI policy on the ground suggests that some corrective actions have not been taken, or implementation plans are still in progress, particularly in regard to PEO (Program Educational Objectives), Course Mappings, and Taxonomy Levels.	The survey forms used for evaluating PEO attainment have been revised. Course Learning Outcomes (CLOs), along with their respective domains and taxonomy levels, have been updated and are subject to continuous review at the beginning and end of each semester. Additionally, corrective actions have included the revision of the internship evaluation process. The department is committed to continuously improve the processes and train the faculty related to these processes.
	Only partial actions have been taken, and the implementation plans are unsatisfactory. The concerns and weaknesses identified in the last accreditation visit report, particularly in Criterion 2, 3, and 8, still require comprehensive and dedicated attention.	The Department has strengthened its Continuous Quality Improvement (CQI) processes for the assessment and attainment of Program Learning Outcomes (PLOs) under Criterion 2. Key improvements include the implementation of an autonomous internship feedback system and the redesign of the Graduate Exit Survey to eliminate ambiguities and replace direct self-assessment questions with outcome-oriented indicators. To further enhance the effectiveness of Outcome-Based Education (OBE), faculty members are provided with regular training and capacity-building sessions. Course Learning Outcomes (CLOs) are systematically reviewed at the beginning and end of each semester, while CLO–PLO mappings are periodically evaluated and revised where necessary to ensure their continued relevance and alignment with program objectives. The Departmental Continuous Quality Improvement (CQI) Committee conducts a comprehensive analysis of assessment results and stakeholder feedback, identifies areas requiring improvement, and recommends evidence-based corrective actions to support the continuous enhancement of academic quality and program outcomes.

	There is a significant number of journal publications, and senior faculty members are actively involved in funded research and development projects. However, it's noteworthy that only one faculty member is engaged in consultancy activities.	The department is committed to providing faculty members with opportunities to enhance their research profiles. These initiatives include, but are not limited to, faculty placement programs, financial support for research activities, and encouragement and incentives for the development of spin-off ventures. Several faculty members are involved in consultancies. Details are presented in Criteria 9
	A limited number of new initiatives have been undertaken. Moreover, the implementation of OBE (Outcome-Based Education) requires improvements in the assessment and evaluation processes.	The Department has undertaken numerous initiatives to strengthen the implementation of Outcome-Based Education (OBE) and continuously enhance its assessment and evaluation processes. These initiatives include the introduction of an autonomous internship feedback system, revision of indirect assessment instruments, regular OBE training and capacity-building sessions for faculty, periodic review of Course Learning Outcomes (CLOs) and CLO–PLO mappings, and systematic analysis of assessment results through the Departmental Continuous Quality Improvement (CQI) Committee. To ensure the quality and consistency of assessments, all assessment instruments are reviewed by the respective Knowledge Group Heads to verify their alignment with the defined Course Learning Outcomes (CLOs), appropriate cognitive levels, and overall assessment quality. This review process helps ensure that assessment practices remain consistent with the principles of Outcome-Based Education. Assessment data and attainment results are analyzed at the end of each semester, and evidence-based corrective actions are identified and implemented wherever required. This structured review and continuous improvement mechanism has significantly strengthened the Department's OBE implementation and supports the ongoing enhancement of assessment and evaluation processes.

9. Industrial Linkages	Both formal and informal mechanisms are in place; however, there is a need for the assessment tools and methods to have a stronger correlation with the Program Educational Objectives (PEOs).	To strengthen the assessment process, the collection of employer feedback has been institutionalized through a dedicated Online Internship Feedback Module integrated with the Department's existing Thesis Management System (TMS). This system facilitates systematic collection, storage, and analysis of employer evaluations for every internship. It also supports the generation of consolidated, batch- wise reports with complete employer responses, thereby enhancing data reliability, traceability, and evidence- based Continuous Quality Improvement (CQI) to align with Program Educational Objectives.
	There are projects that are jointly supervised by the industry, but there is no substantial evidence of sponsored design projects from the industry.	The department acknowledges the observation. While several Final Year Design Projects (FYDPs) are jointly supervised by industry professionals, formal industry-sponsored design projects have been limited. The department is actively strengthening collaborations with industry partners to secure sponsored design projects through MoUs, funded industry engagements, and collaborative R&D initiatives. These efforts are expected to increase the number of industry-sponsored projects in upcoming academic sessions. The details are attached as **Annex XXXX**
	Taking into account the overall faculty strength, there is limited faculty involvement observed in engaging in projects and providing consultancy services to the industry.	The department acknowledges the observation and has initiated measures to enhance faculty engagement with industry. Faculty members are being actively encouraged to undertake consultancy assignments, collaborative R&D, and industry-sponsored projects through strengthened partnerships with industry, dedicated facilitation mechanisms, and incentive-based initiatives. These efforts are expected to significantly increase faculty participation in consultancy and applied industrial projects in the coming years. The details are attached as **Annex XXXX**